

Table 3.1: Thermal Resistances for CSR Hebel Products

CSR Hebel Product		Nominal Dry Density (kg/m ³)	Dry State Thermal Conductivity (W/m.K)	Product Thickness (mm)	Dry State Thermal Resistance R _{dry} (m ² .K/W)	5% M.C.* Thermal Resistance R _{5MC} (m ² .K/W)	10% M.C.* Thermal Resistance R _{10MC} (m ² .K/W)
Block	Thermoblok	470	0.11	100	0.86	0.71	0.61
				150	1.29	1.07	0.92
				200	1.72	1.43	1.23
				250	2.15	1.79	1.54
				300	2.58	2.15	1.84
	Sonoblok	650	0.16	100	0.60	0.50	0.43
				150	0.91	0.76	0.65
				200	1.21	1.01	0.86
				250	1.52	1.26	1.08
				300	1.82	1.52	1.30
Panel	PowerPanel™	510	0.12	75	0.59	0.49	0.42
				75	0.59	0.49	0.42
	SoundFloor™	510	0.12	75	0.59	0.49	0.42
				75	0.59	0.49	0.42
	Wall (Sound Barrier)	550	0.13	100	0.72	0.60	0.51
				150	1.09	0.90	0.77
				200	1.45	1.21	1.03
				250	1.81	1.51	1.29
	Floor	580	0.14	300	2.18	1.81	1.55
				100	0.68	0.57	0.49
				150	1.03	0.85	0.73
				200	1.37	1.14	0.98
Lintels		650	0.16	250	1.71	1.43	1.22
				300	2.06	1.71	1.47
				100	0.60	0.50	0.43
Stair Treads		650	0.16	150	0.91	0.76	0.65
				200	1.21	1.01	0.86
				175	1.05	0.88	0.75

Note: R-Values for different thickness can be interpolated and extrapolated from the values in the table. The R-Values for moisture contents within the Dry State to 5% moisture content, 5% to 10% moisture content ranges, can be linearly interpolated. For moisture contents exceeding 10%, contact CSR Hebel.
M.C. denotes moisture content of AAC.

Thermal Resistance of Air Spaces, R_{as}

When accurately calculating the total thermal resistance of a building component the thermal resistance of air spaces needs to be considered.

Thermal Resistance of Air Spaces, R_{as}
The greater the value,
the better the thermal insulation

Thermal Resistance of Surface Air Film, R_s

When calculating the total thermal resistance of the building component between internal and external environments, the thermal resistance of the air layers adjacent to the surfaces, R_s must be considered.

Thermal Resistance of Surface Air Film, R_s
The greater the value,
the better the thermal insulation

Thermal Transmittance, U

The thermal transmittance, U is a direct measure of the thermal insulating ability of a given building component air to air. It is obtained by reciprocating the total thermal resistance of the building component, R_T (i.e., U = 1 / R_T).

Thermal Transmittance, U
The lower the value,
the better the thermal insulation